

UNIT 3: Motion in a Straight Line (Kinematics)

Subject: Physics

Grade: Grade 9

Position, displacement, speed, velocity, acceleration, uniform motion, and motion graphs.

Position, Distance, and Displacement

Motion is a continuous change in position relative to a reference frame. A body is at rest if position does not change with time in that frame, and in motion if position changes.

For one-dimensional motion, we usually use the x-axis and origin as a fixed reference point.

Displacement formula: $\Delta S = S_f - S_i$

Distance vs Displacement

Distance is the total path length traveled; displacement is the shortest directed change from initial to final position.

Distance	Displacement
Length of actual path traveled	Shortest directed change between initial and final points

Scalar quantity	Vector quantity
Depends on path	Depends only on initial and final positions
Always greater than or equal to magnitude of displacement	Magnitude always less than or equal to distance

Speed and Velocity

Speed is scalar and describes how fast an object moves: $v = \text{distance}/\text{time}$.

Average speed = total distance / total time.

Instantaneous speed is speed at a specific instant (for very small Δt).

Velocity includes direction:

Average velocity = $\Delta S / \Delta t$

Instantaneous velocity is velocity at a specific moment.

Acceleration and Uniform Motion

Acceleration is the rate of change of velocity and is a vector quantity:

$$a = (v - u) / (t_f - t_i)$$

If velocity decreases, acceleration is negative (deceleration).

Uniform rectilinear motion means constant velocity in a straight line, so acceleration is zero.

Graphical Representation of Motion

Position-time (S-t) graph:

- slope gives velocity

Velocity-time (v-t) graph:

- slope gives acceleration
- area under graph gives displacement (or distance for non-negative speed)

Position (m)	Time (s)
0	0
5	1
10	2
15	3
20	4
25	5
30	6

Speed Limit and Traffic Safety

Speed limits are part of a broader traffic safety system that also includes driver training, road design, vehicle safety features, and law enforcement.

Drivers must follow posted speed limits and adjust speed to traffic and weather conditions.

Worked Examples for Unit 3

Sample applications:

- A cyclist rides 3 km west, then 2 km east. Find distance and displacement.
- A car travels 80 km in 1 hr 30 min. Find average speed in m/s.
- A student walks at 5 m/s for 1.5 hr. Find distance covered.
- Convert 120 km/h to m/s.
- A student has displacement 360 m north in 180 s. Find average velocity.
- A train accelerates from rest to 36 km/h in 20 s. Find acceleration.
- A car changes speed from 7.0 m/s to 12.0 m/s at 2.5 m/s^2 . Find time.